

CS 5330 HW11

1. What makes the problem of object detection and recognition different than image classification?

classification is concerned with just labeling the image, detection also has to care about the location of the object in the image

2. How do the R-CNN style networks decide what parts of the image to analyze?

It uses one deep network to suggest region proposals (area where a potential bounding box goes) and runs different CNN on those regions to determine the object label

3. How do the YOLO style networks decide what parts of the image to analyze?

YOLO divides the image into a grid, and predicts the bounding box (width + height) and class for an object centered on that grid

4. How does a Transformer decide what parts of the image have objects in them?

The object queries in the transformer network each learn a specific part or feature of the image during training. Each query then outputs one prediction (bounding box + class) at the end of training. The part of the image that has an object in it will be the object which a specific query thought was most "important"

5. Why does a Transformer not need to do non-maxima suppression of its outputs?

Because by nature each query only matches to one object

6. When doing transfer learning, what part of the deep network is modified to adapt to the new problem?

Only the final output layer is modified

7. What is the purpose of the deep network when doing a matching problem, such as face recognition?

The purpose is to compute an embedding for each image, which is then compared to other embeddings of other images via a distance metric.

8. What methods are used to do the actual matching of an input to a library of examples?

Distance metrics such as sum-squared distance, cosine distance, etc.

9. What is the purpose of contrastive learning?

This compares the two image pairs. If a good match, it tries to make those images "closer" together in the embedding space. The opposite is true for a bad match

10. In the contrastive learning loss function, what is the purpose of the margin?

In the contrastive learning loss function, bad matches only are forced apart if they have a loss value that is past some margin

11. Why is it harder to find good negative pairs when doing contrastive learning?

Most bad matches do not tell anything useful, only those pairs that share some similarities, but have bad matches, can really inform the model